

Game Theory: Environmental Applications

Business Economics, Climate Change and the Environment

Simon Quinn

Module outline

LECTURE	DATE	TOPIC
1	9 September	Growth, Transformation and Transition
2	11 September	Introduction to Game Theory
3	16 September	Game Theory and Auctions
4	17 September	Game Theory and the Environment
5	23 September	Academic Papers Explained
6	25 September	Academic Papers Explained
7	30 September	Consumer Preferences
8	2 October	Institutions, Power and Inequality
9	7 October	Supply and Demand
10	9 October	Market Successes and Failures

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1. We will return to **fairness** in **Lecture 8**; and
2. We will discuss '**cap-and-trade**' policies in **Lecture 10**.

The 'uniform price auction'

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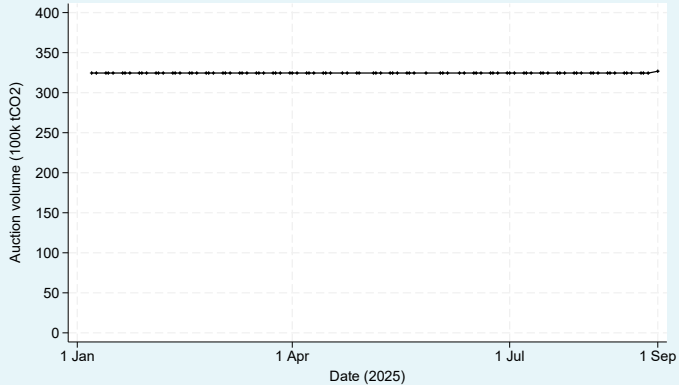
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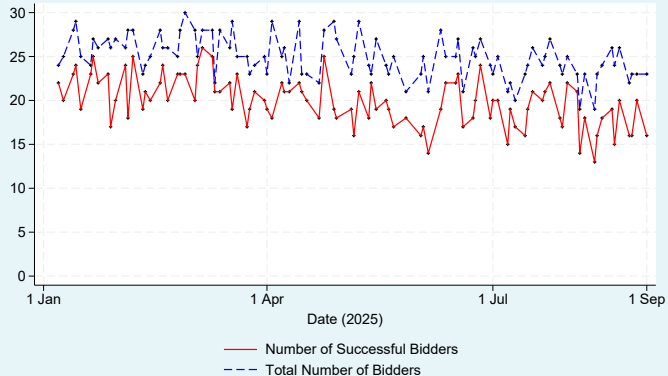
1. Understand the mechanism, using an example; and
2. Discuss an optimal bidding strategy under a uniform price auction.

Recent ETS auctions: **volume**



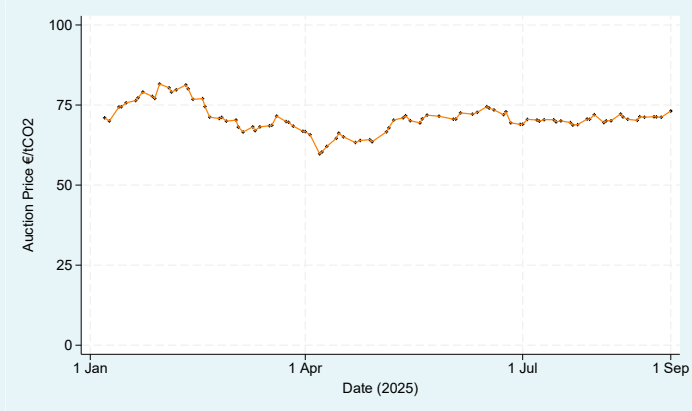
<https://www.eex.com/en/market-data/environmental-markets/eua-primary-auction-spot-download>

Recent ETS auctions: **number of bidders**



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Recent ETS auctions: **price**



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Recent ETS auctions: **total revenue**



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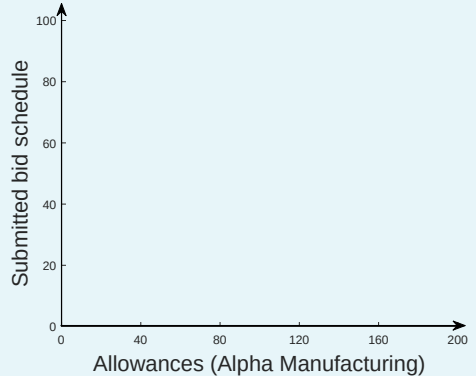
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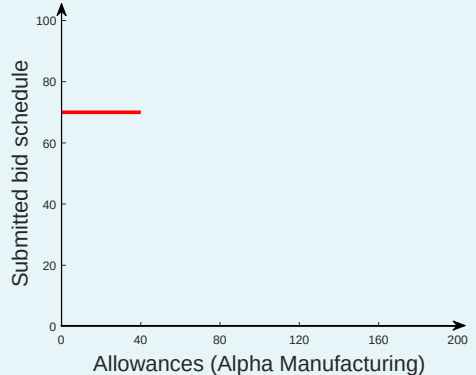


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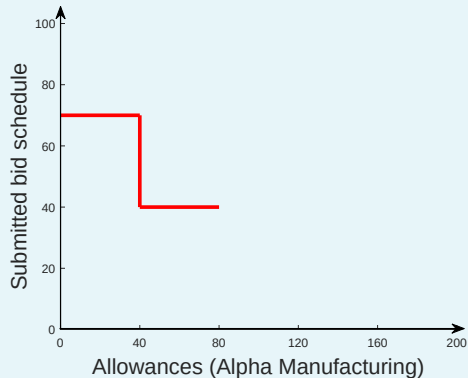


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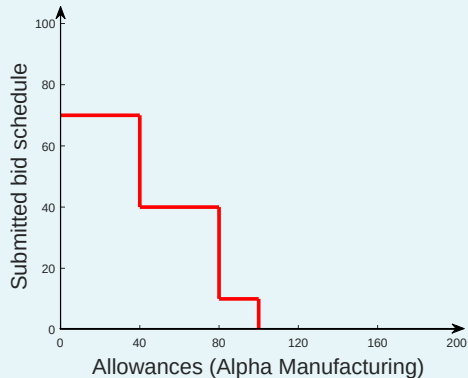


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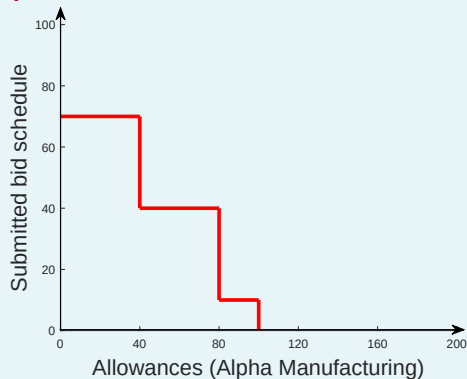


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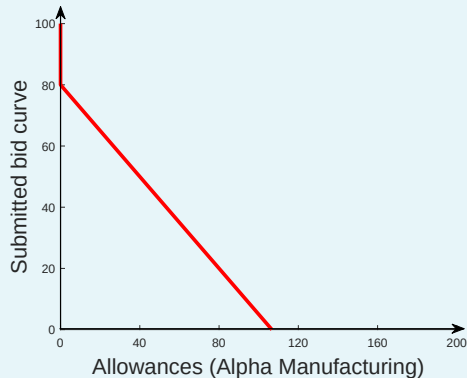
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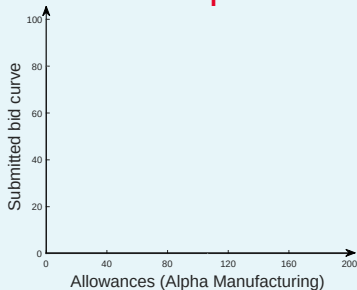
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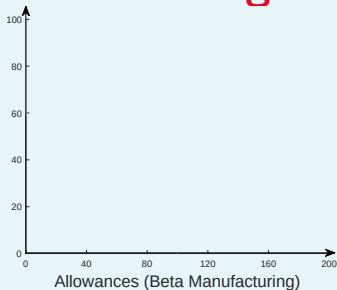


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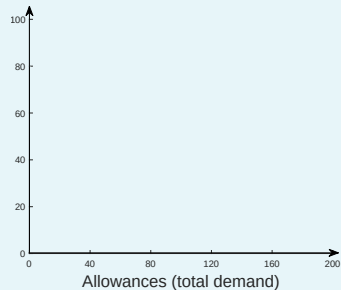
Uniform price auction: **Allocating allowances**



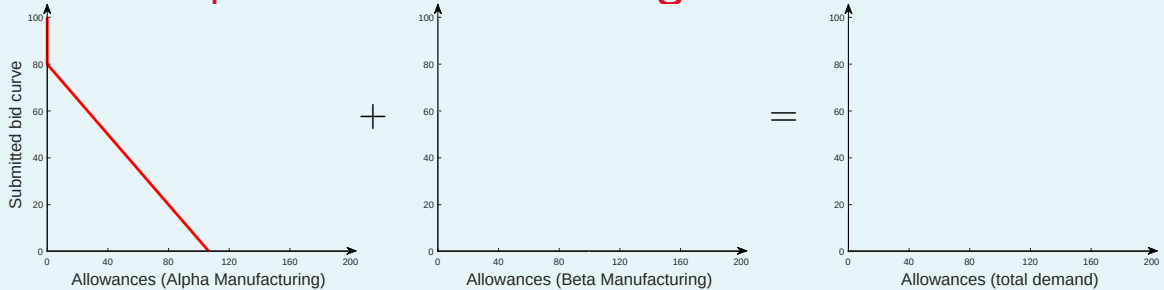
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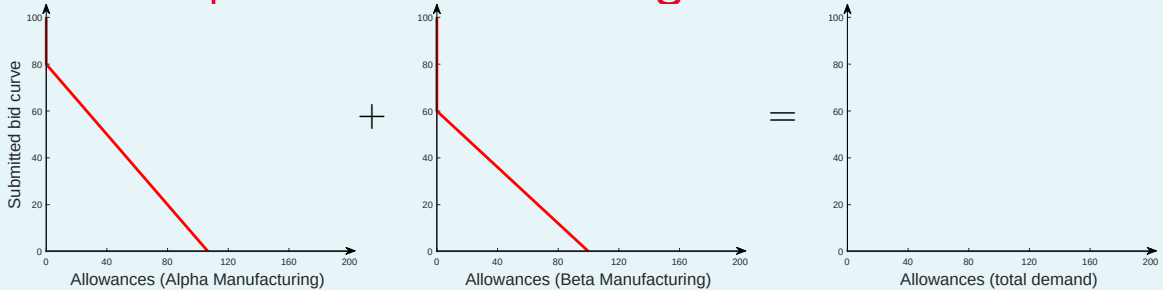


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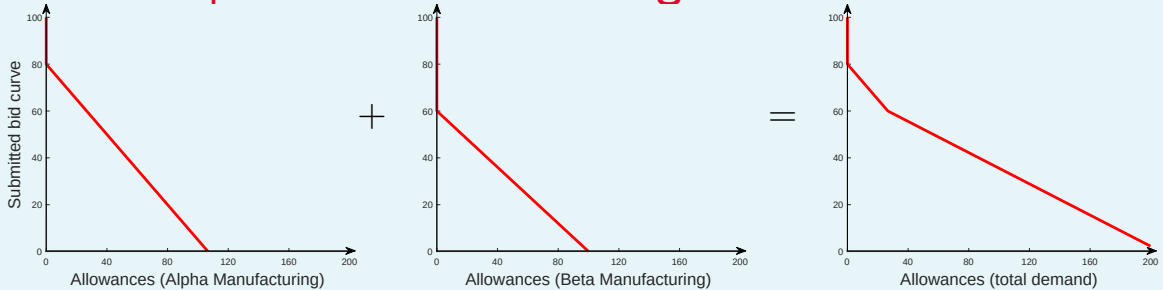
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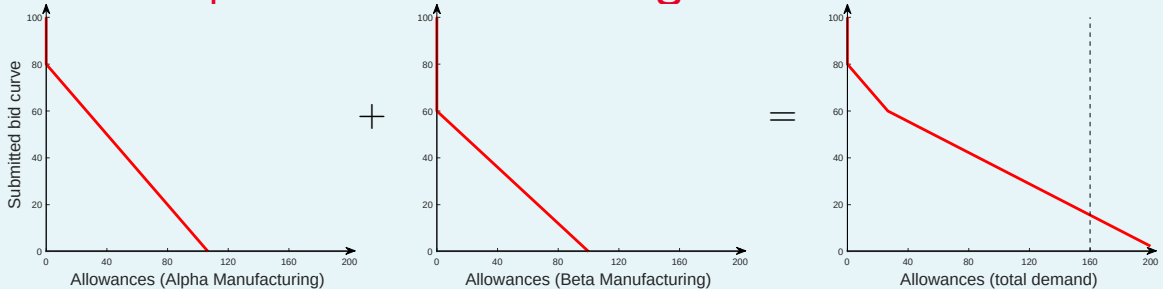
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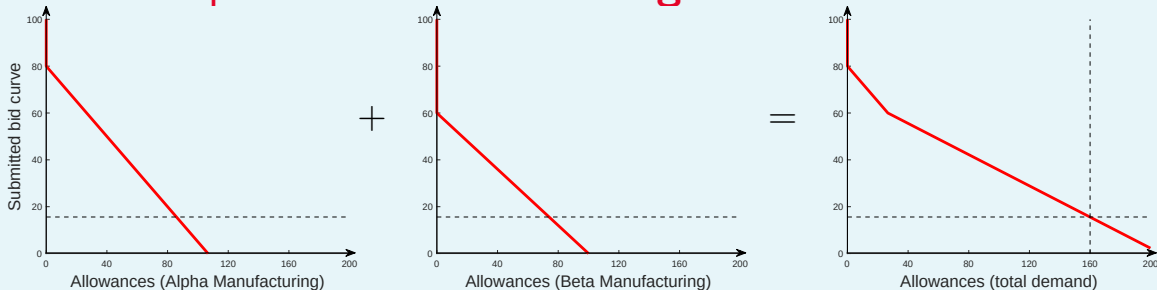
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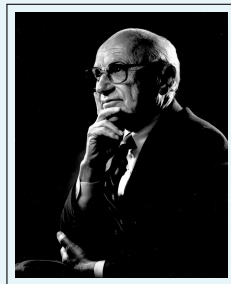


1. Add the total demand, for each possible price;
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3. Apply that price to each bidder, to decide individual allowances.

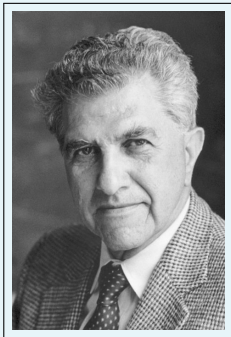
Two (**incorrect**) claims about uniform price auctions...

A [uniform-price] auction proceeds precisely as [a discriminatory auction] with one crucial exception: All successful bidders pay the same price, the cut-off price. An apparently minor change, yet it has the major consequence that no one is deterred from bidding by fear of being stuck with an excessively high price. **You do not have to be a specialist. You need only know the maximum amount you are willing to pay for different quantities.**

Milton Friedman (Nobel Prize in Economics, 1976)
Wall Street Journal, August 1991



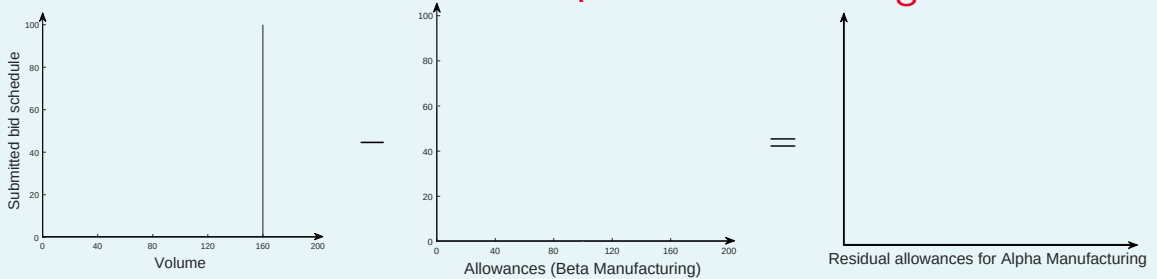
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All of that [bid-shading] is eliminated if you use the [uniform-price] auction. **You just bid what you think it's worth.**

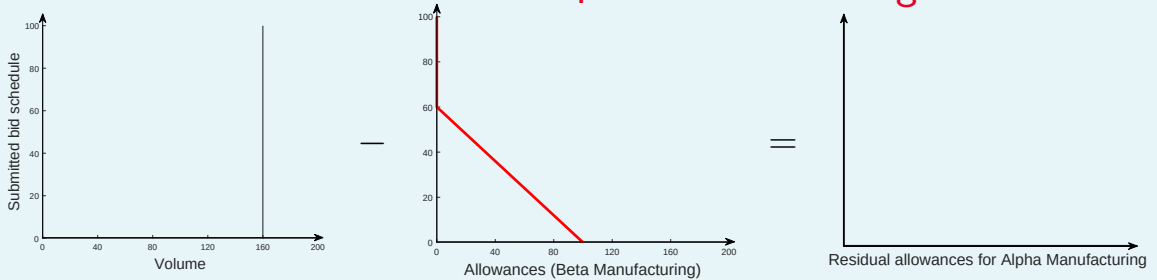
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Residual allowances for Alpha Manufacturing



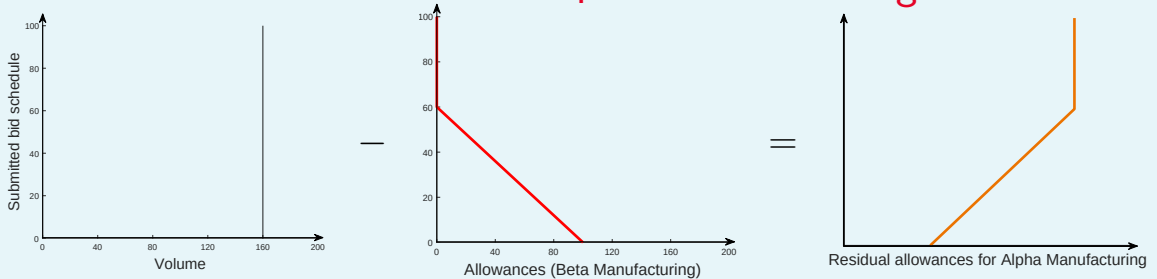
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3. For each price, subtract Beta's bid curve from the volume cap, to calculate the '**residual allowance**' for Alpha Manufacturing.

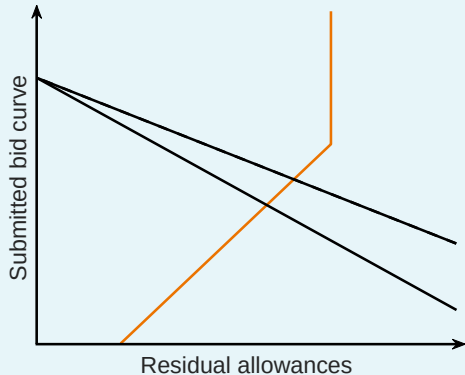
Bid shading in a uniform-price auction



Suppose that Alpha Manufacturing has decided its **valuation** for each available allowance.

Should Alpha **bid its valuation**?

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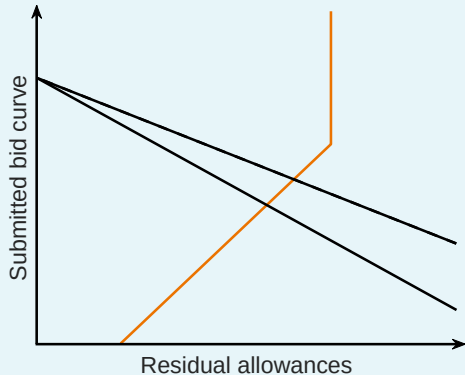


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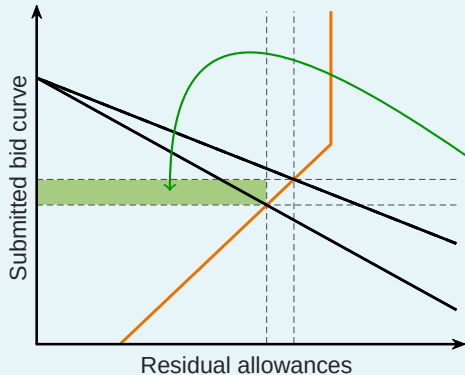
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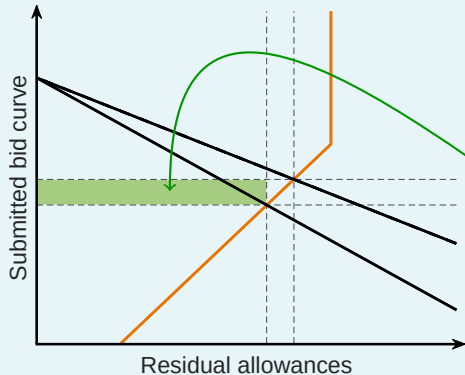
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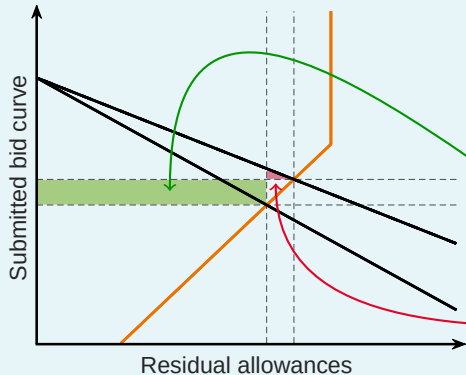
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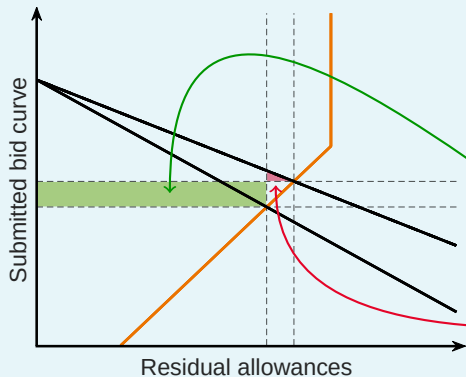
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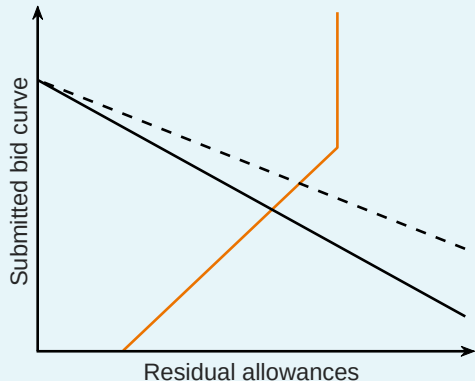
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Clearly, the **gain** from bid-shading in this uniform-price auction exceeds the **loss**...!

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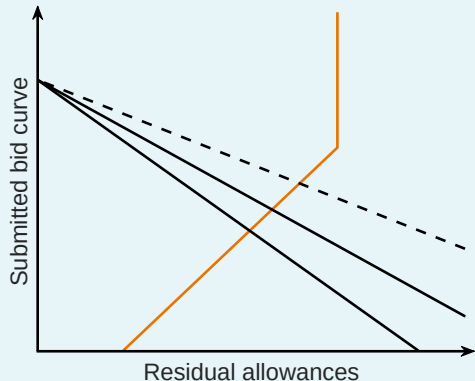


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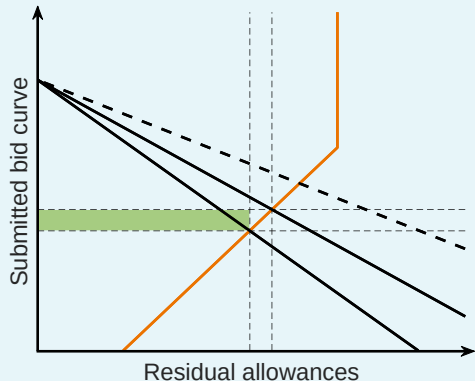


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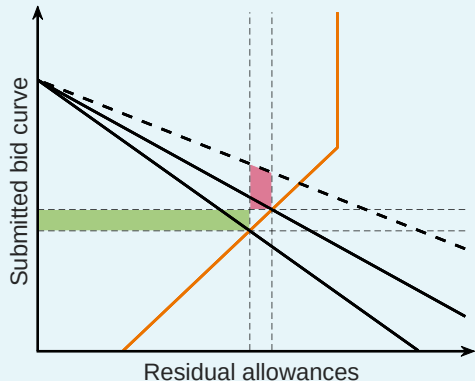


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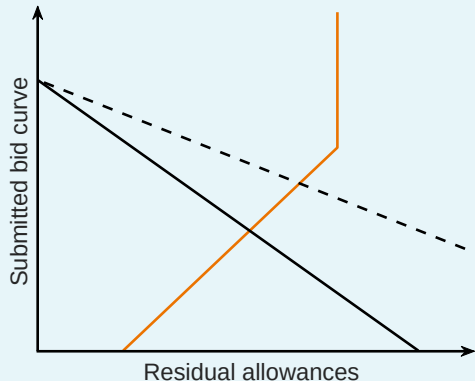


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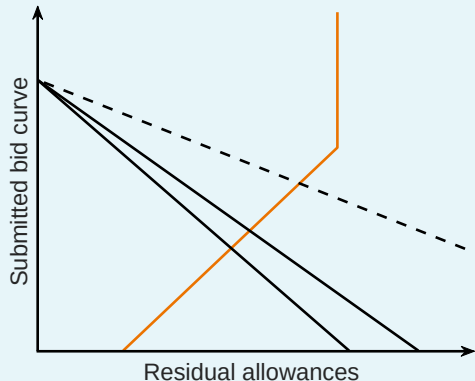


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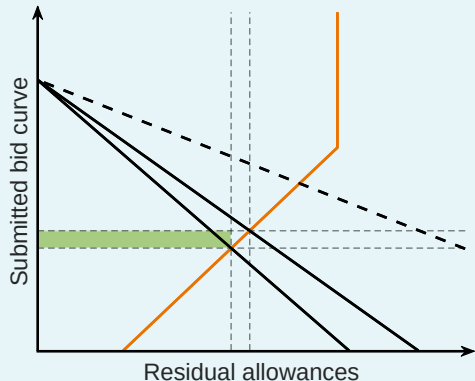


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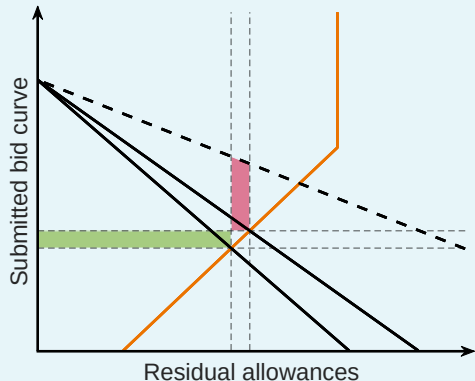


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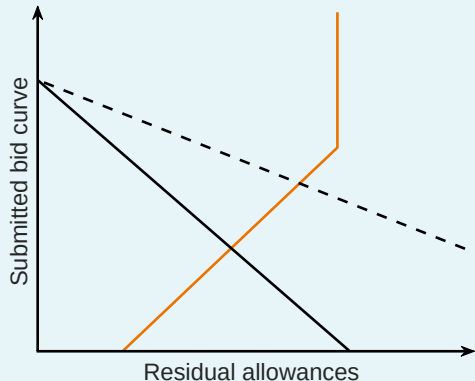


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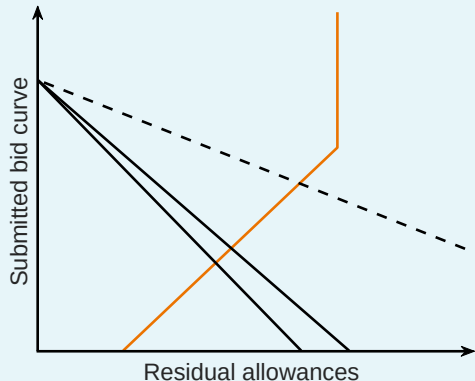


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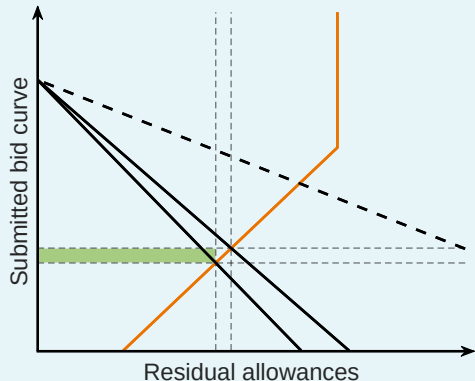


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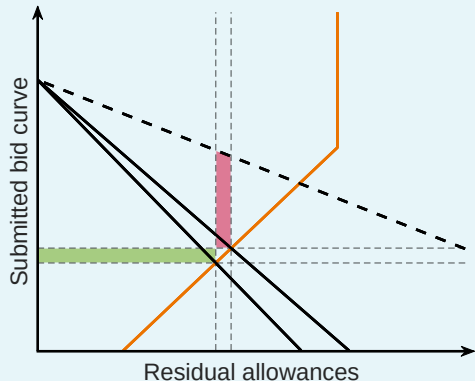


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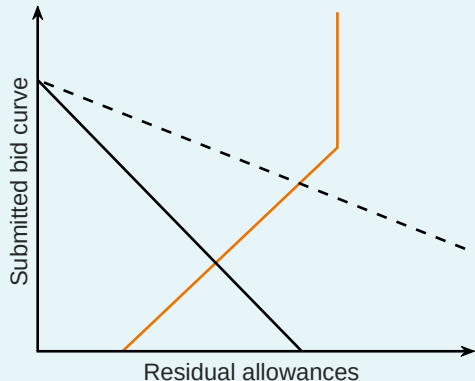


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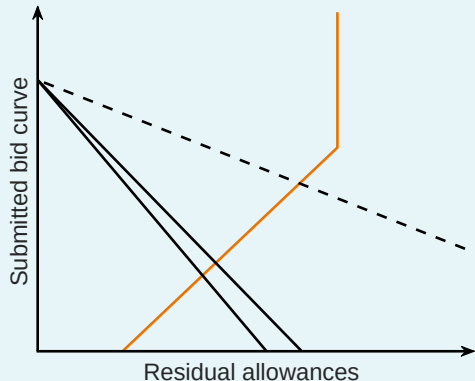


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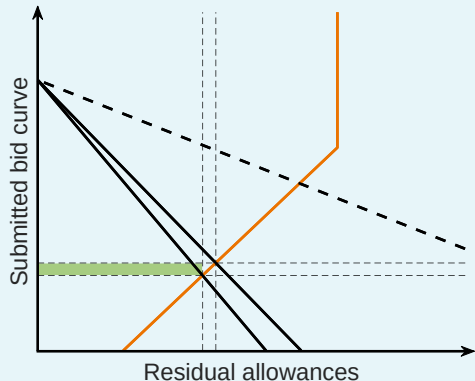


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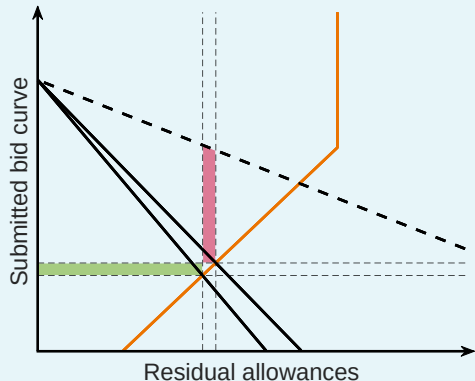


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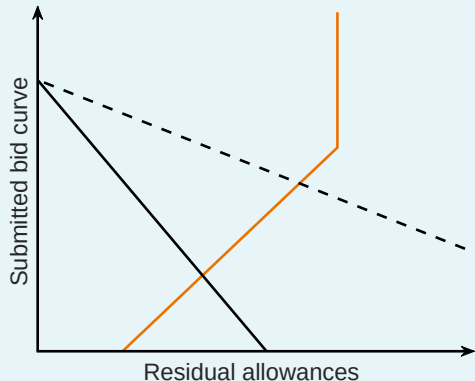


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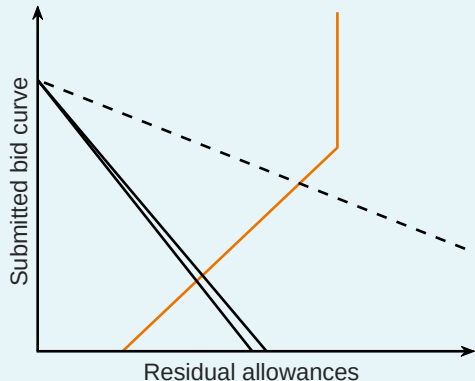


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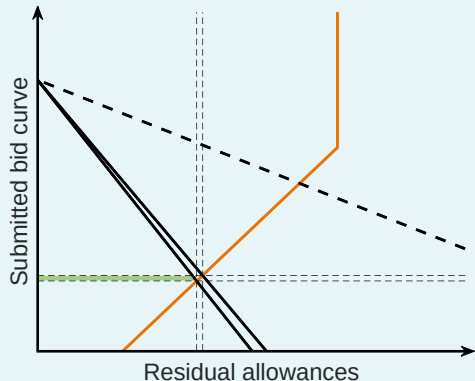


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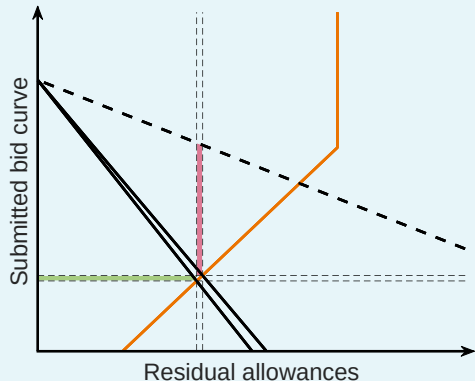


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Key concept: Marginal trade-offs

The **marginal gain** from bid-shading equals the **marginal loss**.

A thought experiment. . .

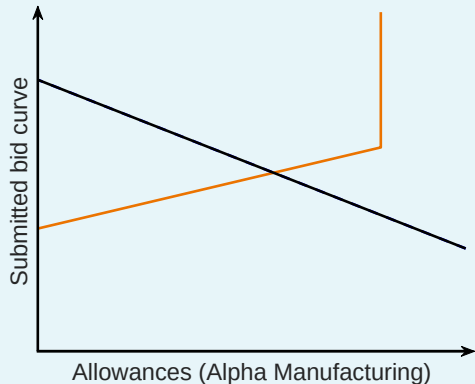
What if Alpha Manufacturing were **tiny**, compared to Beta Manufacturing? (Or, indeed, if there were **many other bidders**, besides Beta?)

A thought experiment. . .

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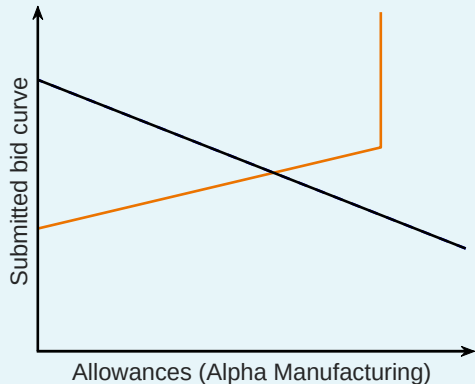


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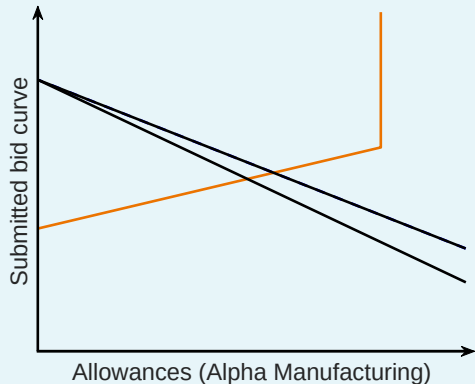


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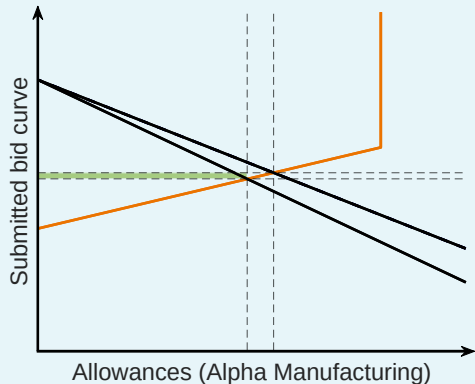


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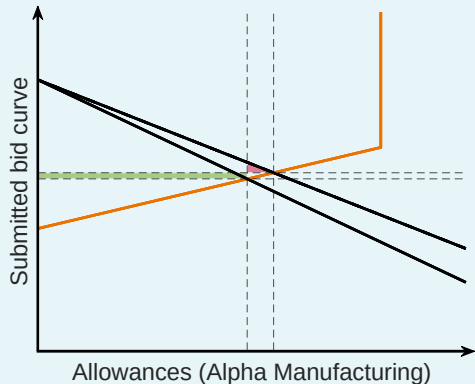


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2. We discussed how this kind of auction is **implemented**.
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Fairness and efficiency in games

Key concept: Ultimatum game

An **ultimatum game** is a strategic setting in which one player (the Proposer) offers a share of a prize to a second player (the Responder). If the Responder accepts, the proposal is accepted. If the Responder rejects, each player receives nothing.

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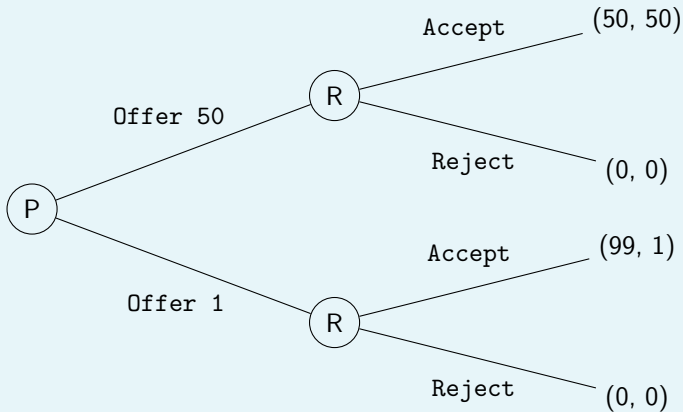
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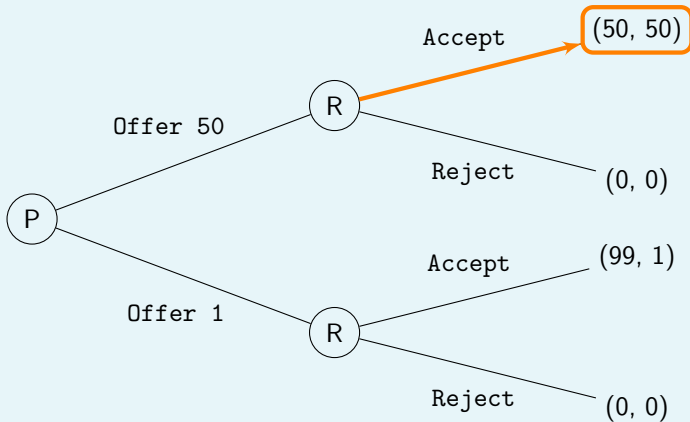
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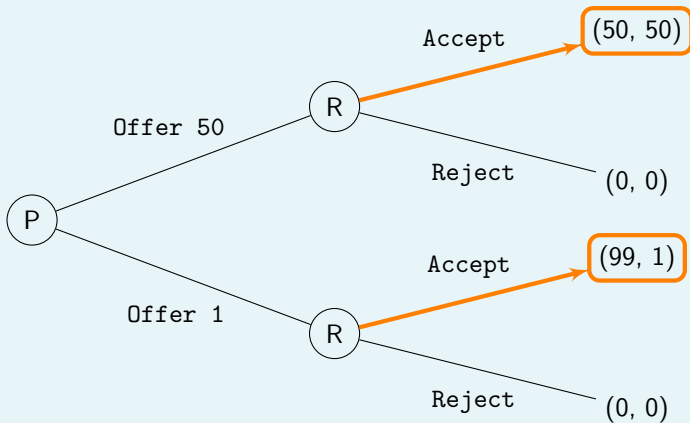
An 'ultimatum game' with two possible offers



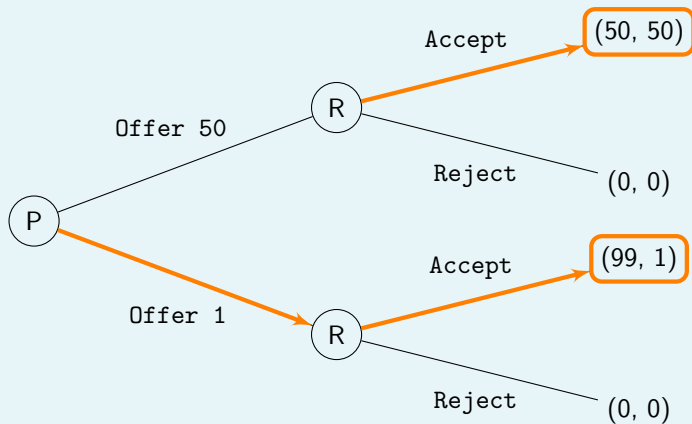
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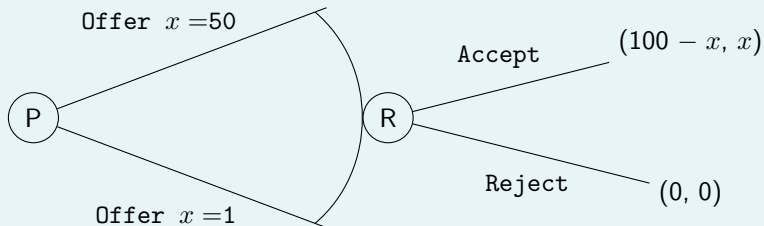
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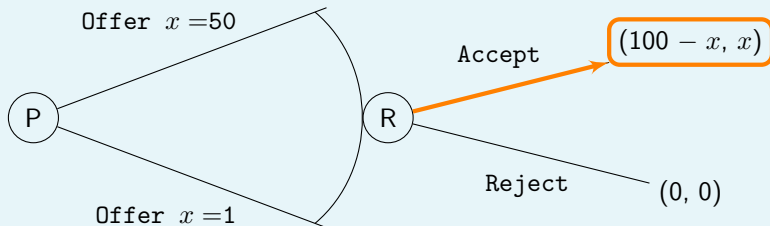
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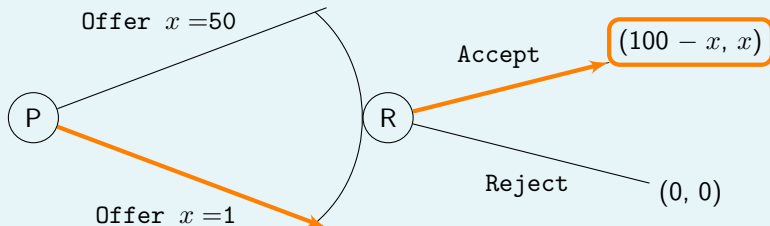
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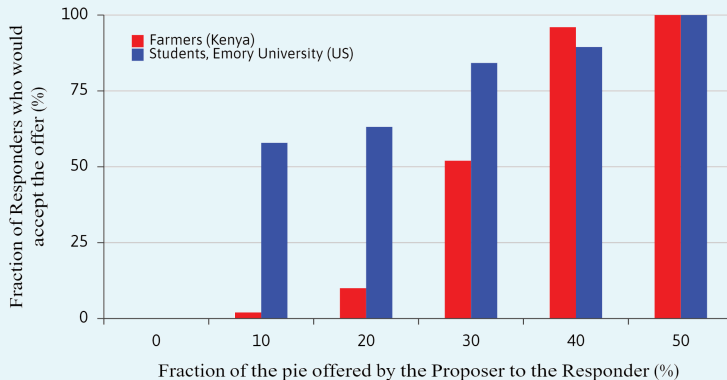
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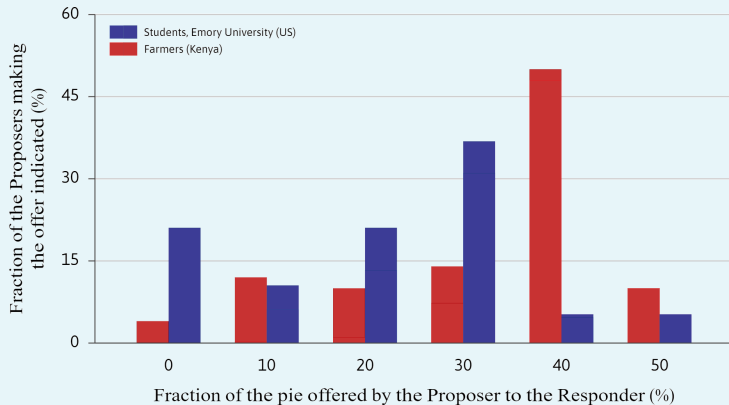
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If you were the Proposer, what offer would you make?

Responder behaviour in two field settings



Proposer behaviour in two field settings



Two questions...

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In your opinion, what is the ultimatum game really testing? Is this about inequality? Procedural fairness? Substantive fairness?

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Key concept: Pareto efficiency

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Question

In the ultimatum game, which outcomes are Pareto efficient? (And what does this tell us about the notions of **fairness** and **efficiency**?)

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4. Outcomes can be Pareto efficient and very unfair – and vice versa.

Game Theory: Environmental Applications

Business Economics, Climate Change and the Environment

Simon Quinn